ANTAR

Intelligent Conversational AI & Crime Analytics Platform

for Karnataka State Police · State Crime Records Bureau

End-to-End Guide — Operational and Technical

Datathon 2026 · Final-day reference · 21 September 2026

10,00,000	1,155	0.66 s	903	EN + ಕನ್ನಡ
FIRs searchable	police stations	mean answer time	automated tests	text and voice

Contents

1 Executive summary 2
2 The problem and the people 3
3 Antar's constitution 4
4 Operational: an investigation, end to end 5
5 Operational: the screens 6
6 Operational: asking Antar and reading answers 7
7 Technical: architecture 8
8 Technical: how a question becomes an answer 9
9 Technical: data, scale and performance 10
10 Technical: the intelligence agents 11
11 Governance, security and trust 12
12 Requirements coverage and measured impact 13
13 Final-day kit 14

1 Executive summary

The problem. Karnataka's State Crime Records Bureau holds crime records from 1,100+ police stations, but officers reach them through static dashboards and manual queries. A simple question — *which district has the most kidnapping cases, and is it rising?* — needs a report request, a spreadsheet and time an officer under pressure does not have. Nothing links cases, offenders, places and money, nothing warns early, and nothing answers in Kannada.

What Antar is. An analyst colleague that reads the FIR registry so officers don't have to. An officer asks in plain English or Kannada — typed or spoken — and gets a direct answer in under a second, with the evidence, the FIRs it came from, what it cannot see, and a next step. Behind the chat sit the Brain (full-dataset analytics and crime cycles), the Map, Predict, Search, and an Approvals flow where a supervising officer signs off any real-world action.

Why it is different

- **It never answers a different question.** If Antar cannot apply part of a question — an unknown crime type, a time word, a place without the data — it says so. It never quietly widens the search and returns a confident wrong number.
- **Every number is traceable.** The language model never writes database queries; it picks from 19 reviewed query templates. Every answer carries an evidence class — Known, Derived, Inferred or Unknown — and its source FIRs.
- **The officer decides.** Antar recommends; patrol deployments and watchlist changes go to a supervising officer for approval, and every step is written to a tamper-evident audit chain.
- **It keeps working without the LLM.** A deterministic answer writer handles every question type, so answers stay fast and correct when the cloud model is slow or offline.

Measured impact

What	Before	After
Searchable FIR registry	4,25,022 rows (part cloned)	**10,00,000** — 5.75 lakh real KSP records added
Mean time to answer (17 representative questions)	9.4 s (worst 57 s)	**0.66 s** (worst 4.8 s)
FIRs reachable by crime name	33 crime names; 2,32,674 FIRs unreachable	**121 concepts over all 142 crime groups**
Silently dropped filters (wrong-question answers)	Crime types, time words, 'absconding'	**Refused with nearest options**
Full Data Set table, per page	11.2 s	**0.005 s**
Deployment package	1.27 GB database	**84 MB** (indexes self-build)
Automated tests	835	**903**, incl. 51 that enforce Antar's rules

> **Ready for the final:** chat (EN/ಕನ್ನಡ, voice), resolve-or-refuse, evidence classes, Expert console, Brain analytics and crime cycles, absconding accused, network analysis, HITL approvals with audit chain, observability.
>
> **In progress:** beat-level forecasting with measured accuracy (see §12–13) and the Catalyst deployment check.

2 The problem and the people

Designed for officers under time pressure, not for a dashboard audience.

What gets in the way today

Gap	What it costs an officer
Data sits behind reports and SQL	Every question becomes a request to the records room and a wait.
Dashboards show what someone decided to chart	The question actually asked usually isn't on the dashboard.
Cases, people, places and money are not linked	Repeat offenders and organised networks stay invisible across stations and districts.
No early warning	A rising pattern is noticed after it has happened, not while it is forming.
English-only tools	Station staff working in Kannada are left out, or translate by hand.
No evidence trail	A number without its source FIRs cannot go into a report, a briefing or a court file.

Who Antar is for

User	What they need	Where Antar serves them
Crime Records Officer (SCRB / DCRB analyst) — *primary*	Exact counts, rankings, trends and case lists they can put in a report without re-checking	Chat, Expert console, Brain, printable report
Station House Officer / station admin	What is happening in my jurisdiction and what to do next — fast	Chat (EN/ಕನ್ನಡ, voice), Map, Predict
Senior officers (SP / DCP / IGP)	A bottom line to act on, with the evidence one tap away	Chat bottom lines, Brain, Predict board
Supervising officer (Approver role)	Review and decide real-world actions others propose	Approvals screen, audit chain

Design principles that follow from this

- **Answer first.** The first line answers the question as asked and can be read alone.
- **Show, don't claim.** Every figure links to FIRs; every judgment says what it rests on.
- **Say what you can't see.** Missing data is reported as missing, never as zero or as a clean record.
- **Both languages, same answer.** A Kannada question reaches exactly the logic its English twin does.
- **Humans act.** Antar proposes; a named supervising officer approves.

3 Antar's constitution

Written in docs/ANTAR_CONSTITUTION.md and enforced in code (app/core/antar_identity.py) — not a prompt.

Identity. Antar (ಅಂತರ್ — *within*) is the intelligence already inside the record: an analyst colleague that is confident about what the records say and plain about what they don't.

The four laws

Law	What it means in practice	Enforced by
1. Never answer a different question	A filter that cannot be applied is refused, never dropped.	Resolve-or-refuse in the query agent
2. Evidence before opinion	Numbers only from reviewed query templates; judgments labelled with their basis.	Template-only SQL; evidence classes
3. The officer decides	Antar informs and recommends; it never acts.	HITL approvals
4. Protect the vulnerable	No victim identity in sexual offences or cases involving children (BNS §72, POCSO §23); no targeting individuals by caste, religion or community.	Redaction on every answer; Vigil screening

Evidence classes (aligned with ICD 203 analytic standards)

Class	Meaning	Example
K — Known	Read directly from a record	5,950 kidnapping FIRs registered in Bengaluru City
D — Derived	Calculated from records: shares, rates, rankings, trends	37% of the Karnataka total; down 9%
I — Inferred	Antar's judgment, in calibrated words, with its basis	Likely (55–80%) to stay highest next quarter
U — Unknown	A gap — the data is absent or unrecorded	Arrest counts aren't recorded for these FIRs

Voice and answer template

- Facts are stated as record; "I" appears only for limits and judgments. No salutations, no filler, exact numbers, calm register.
- Kannada: formal administrative Kannada; FIR, BNS/IPC sections, NDPS, POCSO, place names and numbers stay as on the FIR form.

> **Bottom line** — one sentence that answers the question, with the key number.
>
> **Context** — one line: direction, comparison or scope. **Evidence** — ranked list or table (≤ 10 rows).
>
> ▸ **Sources** ▸ **Limits** — one tap away. **Next** — one next step, plus 2–3 follow-ups checked against the data.

4 Operational: an investigation, end to end

The demo story — a Crime Records Officer from question to approved action.

#	Step	Where	What the officer does → what Antar returns
1	**Orient**	Chat	*"Which districts have the most kidnapping and abduction?"* → Bengaluru City leads — 5,950 FIRs, 37% of Karnataka, followed by Bengaluru Dist (953) and Belagavi Dist (640); steady within ±5%. Badge: Derived.
2	**Narrow**	Chat	*"Absconding accused in Bengaluru City"* → 66,576 live cases with accused still at large, each FIR showing "4 of 4 accused not arrested".
3	**Verify**	Expert console	Opens side by side (50/50). Brief with every claim's evidence class, the cited FIRs, and the agent trace with timings and integrity hash.
4	**Connect**	Chat · Network	Suspect and identifier questions (shared vehicle, UPI ID, IP, accused ID across districts) → linked FIRs and the network graph with its most central members.
5	**Pattern**	Brain	Full Data Set (10 lakh FIRs) and the crime-cycle clock: seasonality by crime category, by year and month, e.g. crime peaking March–August over eight complete years.
6	**Anticipate**	Map · Predict	Hotspot significance (Gi*) and the prediction board with ranges and reasons; a beat-level forecast is being finalised (§12).
7	**Propose**	Chat / Map → Approvals	Officer proposes a patrol redeployment with a written justification. Vigil screens it for prohibited targeting.
8	**Decide**	Approvals	A supervising officer (Approver role — never the proposer) reviews the exact, hash-pinned proposal and approves or rejects with a reason.
9	**Record**	Audit · Export	Every step is on the hash-chained audit ledger; the session exports as a printable report (Print → Save as PDF).

Demo accounts

Account	Role	Does in the story
U-7001 — Operations Officer	Admin (operations)	Asks questions, works the case, proposes the patrol (steps 1–7)
U-9001 — Supervising Officer	Approver	Reviews and approves or rejects the proposal (step 8); cannot approve own proposals

Sign in with the **[OPERATIONS ACCESS]** or **[APPROVER ACCESS]** shortcut on the login screen, then Authenticate. Keep one window per account so the hand-off in steps 7–8 is live.

> **Where Antar says no — and why that builds trust.** *"Absconding criminals in Indiranagar"* → "Accused and arrest counts aren't recorded for the 205 FIRs in Indiranagar … this is missing data, not a clean record." *"How many burgalry cases"* → "I could not match 'burgalry'. Closest: Burglary, Burglary – Day."

5 Operational: the screens

Screen	What it is for	Key things to use
Chat (Antar)	Ask anything about FIRs in English or Kannada, typed or spoken.	Mic for voice · EN | ಕನ್ನಡ toggle · answer card with badge, Sources, Limits, Next · Listen, Copy, Regenerate, Export, feedback
Expert console	Analyst view beside the chat — switch on Expert mode.	Brief (bottom line, evidence by claim K/D/I/U, ranking table) · Evidence (cited FIRs) · Trace (agent steps) · Notes · Operations queue (pattern scan, network analysis, threat assessment, OSINT)
Brain	The whole registry at a glance.	Full Data Set: KPIs, crime categories, yearly trend, top districts, searchable FIR table (ACC-id search) · Crime-cycle clock with year and month filters · Intelligent Data constellation
Map	Where crime concentrates.	District choropleth and drill-down · Gi* hotspot rings (statistically confirmed clusters) · heat · dark zones (where the least is known: data gaps, reporting silence, unresolved cases) · anchor areas (likely home ground of an unidentified offender)
Predict	What may come next, with reasons.	Prediction board ranked by risk · dashboard · scenario ranges · reasoning per zone · model card
Search	Find people and identifiers across FIRs.	Name / alias / phone / vehicle search, tolerant of spelling mistakes · profiles and linked cases
Approvals	Human-in-the-loop decisions.	Pending proposals with justification and Vigil screen · approve / reject with reason · expiry · pending-count badge in the nav
Docs · About · Account	Guides, product story, officer session.	Operations and Approver demo accounts

Everyday conveniences

- **Answer-first scrolling:** each new question pins to the top of the screen and the answer streams beneath it.
- **Listen to answer:** spoken in English or Kannada; shows a spinner while preparing, a stop control while playing; replays are instant.
- **Follow-ups remember context** within a session, and every suggested question is checked against the data before it is offered.

6 Operational: asking Antar and reading answers

Ask for…	English example	Kannada example
Counts	How many murders in 2023?	ನಿನ್ನೆ ಕಳ್ಳತನ ಎಷ್ಟು
Rankings / hotspots	Which district has the most narcotics cases?	ಕಳ್ಳತನ ಎಲ್ಲಿ ಹೆಚ್ಚು
Trends	Theft trend in Mysuru City	ದರೋಡೆ ಪ್ರವೃತ್ತಿ ತೋರಿಸಿ
Crime mix of a place	Which crime is most common in Mysuru City?	—
Case lists / recent	Recent crimes in Belagavi Dist this week	—
Absconding accused	Absconding accused in Bengaluru City	ಬೆಂಗಳೂರು ನಗರದಲ್ಲಿ ತಲೆಮರೆಸಿಕೊಂಡ ಆರೋಪಿಗಳು
Outcomes	Conviction rate by district	ಜಿಲ್ಲಾವಾರು ಶಿಕ್ಷೆ ಪ್ರಮಾಣ
Law split	IPC/BNS vs special and local laws	—
Identifiers & people	FIRs linked to vehicle KA01AB1234 · history of ACC-KA-110413	—
Patrol planning	Beat hotspots in Bengaluru City	—

What Antar understands

- **Time:** today, yesterday, this week, last 7 / 30 days, last N weeks / months / years, this / last month or year, a named year.
- **Crime names:** 121 concepts covering all 142 crime groups in the registry, with everyday words ("burglary" covers day and night burglary; "narcotics" covers NDPS).
- **Places:** all districts and cities in the data; Kannada place names map to the same places.

Reading an answer

- **Bottom line** first, then **context** and **evidence**. The **badge** shows how the headline was obtained: Known, Derived, Inferred or Unknown.
- **Sources** lists the FIRs behind it; **Limits** lists anything missing (e.g. an incomplete month); **How this was answered** shows the steps.
- **Refusals are answers.** If a crime type, period or combination cannot be applied, Antar says so and offers the nearest valid options — it never guesses.

> **Tip for officers:** arrest and accused counts are not recorded for some areas. When Antar says data is missing, choose a district from the Karnataka registry (e.g. Bengaluru City, Mysuru City, Belagavi Dist).

7 Technical: architecture

Layer	Components
Presentation	Next.js 14 · React 18 · Leaflet / MapLibre maps · Recharts · SSE token streaming · bilingual UI (EN / ಕನ್ನಡ)
API	FastAPI (Python) · /api/chat/stream (Server-Sent Events) · map, predict, dataset, voice, HITL, audit, observability routes · role-based access middleware
Orchestration	LangGraph supervisor → fan-out to worker agents → synthesis. Agent fleet declared in an executable manifest (AGENTS.md) that the build verifies against the compiled graph
Identity layer	antar_identity.present(): every answer shaped to the template, evidence-classed, voice-checked and redacted before the first token streams
Intelligence	19 reviewed query templates · self-healing retrieval · Second Brain (dense + sparse hybrid search) · network analysis (NetworkX) · anomaly and hotspot engine · Dwani Kannada agent
Data	SQLite main.db: 10 lakh FIRs + official ER master tables (Act, CrimeHead, CrimeSubHead, District, Unit, Employee, Court, Rank, …) · code-owned query view · 20+ purpose-built indexes
Models	Catalyst QuickML: GLM-4.7-Flash (chat) and Qwen 3.6 35B vision-language · BAAI bge-small-en-v1.5 embeddings (384-d) · ms-marco MiniLM cross-encoder reranker · Bhashini transcription · gTTS speech
Governance	HITL approvals · SHA-256 hash-chained audit ledger · Vigil compliance screen · observability traces and evaluation set

Stack versions

Next.js 14.2.18 · React 18.3.1	FastAPI 0.115.5 · Uvicorn 0.32.1	LangGraph 0.6.11 · Pydantic 2.9.2
NetworkX 3.6.1 · scikit-learn 1.4	sentence-transformers 3.0.1	Leaflet 1.9 · MapLibre 5 · Recharts 3

Deployment

- **Target:** Zoho Catalyst AppSail. The database ships as a data-only **84 MB** archive, expands on first start, and builds its own indexes and query view — so the deployed index layout always matches the code.
- **Resilience:** the port binds immediately; heavy warm-ups (LLM token, analytics caches, every crime-graph year, heaviest queries) run in the background so the first officer never waits.
- **Latency guards:** every LLM call has a wall-clock budget (routing 8 s, composition 6 s, overall 20 s); on timeout the deterministic path answers.
- **Local / GPU path:** the template and model layers are provider-agnostic — the same pipeline runs against a local GPU-hosted model for on-premises deployment.

8 Technical: how a question becomes an answer

Zero hallucination by design — the model never writes SQL and never chooses a number.

#	Stage	What happens
1	**Language**	Kannada detected; a Kannada→English bridge maps analytic terms so both languages reach the same logic.
2	**Route**	LLM selector picks one template and parameters (8 s budget); on timeout, a deterministic router takes over.
3	**Resolve-or-refuse**	Crime names, periods, places and qualifiers (e.g. absconding) must resolve — or the question is refused with nearest options.
4	**Template**	A reviewed, parameterised query — values are bound, never spliced in. Named predicates (e.g. absconding = accused not arrested in a live case) are fixed SQL fragments selected by key.
5	**Execute**	Indexed SQLite; results capped to today; future-dated rows never surface.
6	**Self-heal**	Only when a query returns nothing: repaired template → semantic recall → hybrid BM25 + dense (reciprocal rank fusion) → honest dead-end. Disabled where relaxing would change the question.
7	**Compose**	Deterministic writer (every template, EN + ಕನ್ನಡ) or the LLM (6 s budget) under the constitution's prompt.
8	**Identity**	Template shaping, evidence class, voice check, victim-identity redaction.
9	**Deliver**	Streamed over SSE with layout, citations, chain-of-custody and audit hash.

The 19 officer-facing templates

T1 list by type / place / period	T2 suspect by name or alias	T3 monthly trend	T4 top suspects in a zone
T5 repeat offenders	T7 victim / suspect demographics	T8 counts by investigation status	T9 crimes linked to a suspect
T11 ranking by district / city	T12 most recent crimes	T13 crime-type breakdown	T16 yearly summary
T17 IPC/BNS vs special & local laws	T18 FIRs by vehicle number	T19 identifiers shared across districts	T20 FIRs by UPI ID / IP address
T21 accused history across districts	T22 case outcomes & rates	T23 beat / village hotspots	T14, T15 internal (scan, map)

9 Technical: data, scale and performance

Registry fact	Value
FIRs	**10,00,000** (2016–2026) — 5,74,978 real records added from the 16.7-lakh-row KSP extract, deduplicated on a 14-field signature and sampled evenly across districts and years; existing rows untouched
Coverage	56 district labels (41 normalised districts) · 1,155 stations · 6,384 beats · 4.42 lakh FIRs with coordinates
Schema	Official ER master tables: CrimeHead (99), CrimeSubHead (458), Act (138), Unit (1,056), Employee (15,585), Court, Rank, District, status and gender masters
Crime vocabulary	Read from the registry at start-up: 142 crime groups → 121 concepts, spelling variants merged, families expanded

Performance work (every rewrite verified to return identical rows)

Change	Why it was slow	Result
LLM wall-clock budgets + startup warm-up	Cold LLM call took 75 s; retries had no ceiling	Worst answer 57 s → 4.8 s
COUNT(*) instead of COUNT(column)	Row lookup per row to test for NULL	777 → 99 ms
No UPPER() on date columns	A wrapped column can't use its index	265 → 61 ms
Prefix LIKE → range predicate	Function-call patterns never use an index	5,839 → 158 ms
Expression + covering indexes	Filters and groupings scanned 10 lakh rows	Ranking 798 → 94 ms; crime mix 3,210 → 117 ms
rowid tiebreak on the date index	Each table page sorted the whole registry	11,230 → 5 ms per page
Cached summaries, pre-warmed charts	~17 scans per page visit	10.7 s → instant after first load

Data quality — handled, not hidden

Future-dated rows	Never surface in answers, tables or charts; they unlock automatically on their date.
Spelling variants	"CHAIN_SNATCHING" / "chain snatching", " CYBER CRIME" / "Cyber Crime" merged into one concept.
Truncated last month	The real record ends 10 Mar 2024; that month is excluded from trends and labelled incomplete.
Impossible coordinates	14,512 FIRs carry out-of-state coordinates; the forecasting work flags and excludes them.
Missing arrest data	Reported as missing (evidence class U), never as zero or as a clean record.

Path to crores of FIRs

- Materialised aggregate cubes for dashboards and forecasting; covering indexes for every template; the template layer already renders to Catalyst Data Store (ZCQL) as well as SQLite.
- Local GPU inference for the language model; the deterministic writer means scale never depends on LLM throughput.

10 Technical: the intelligence agents

Twenty agents declared in an executable manifest — the build fails if the documentation and the running graph disagree.

Agent	Role	Agent	Role
Vault (query)	Template queries over the registry	**Link** (network)	Graph builder: PageRank, betweenness, k-core, communities
Motive (profile)	Suspect profiling, repeat offenders	**Sense** (social / pattern)	Demographic insight · MO pattern detection
Lens (geo)	Geospatial intelligence	**Core** (forecast / predict)	Risk forecasting
Beacon (alert_suggest)	Patrol advice from alerts	**Scout** (OSINT)	Open-web scan
Trace (research)	K/D/I/U dossier on a case or subject	**Vault** (brain)	Semantic recall over the Second Brain
Adversary (devil's advocate)	Tries to falsify a hypothesis; base-rate audit	**A5 Vigil**	Bias and compliance screening
Corridor	Cross-district corridor intelligence	**Linkage**	MO-based case matching
Breach (sentinel)	Watchlist tripwires (service)	**Products**	Background product compiler (service)
Supervisor	Intent classification and routing	**Synthesize**	Answer composition, claim verification

Key techniques

- **Second Brain retrieval:** BGE-small embeddings (384-d) for meaning, Okapi BM25 for exact fragments, fused by reciprocal rank fusion, then a cross-encoder reranker — CPU-only and offline-first.
- **Network analysis:** NetworkX over suspects, relationships and shared identifiers (vehicle, UPI ID, IP, accused ID) to surface brokers and organised groups.
- **Anomaly & hotspot engine:** robust MAD z-scores, seasonal residuals, a Bayesian reporting-silence check (possible under-reporting), Getis-Ord Gi* spatial significance, split-conformal ranges and space-time clustering.
- **Dwani:** the Kannada language agent — transcription (Bhashini) in, Kannada answers out, with every number counted from the evidence ledger.
- **Devil's advocate:** challenges a working hypothesis and asks what evidence would disprove it — built-in protection against confirmation bias.

11 Governance, security and trust

Role-based access

Six roles — Constable, Inspector, Analyst, **Approver**, Admin, Superintendent. The Approver ranks above Analyst (to read the evidence a decision rests on) and below Admin (no access to the administrative audit log). Approval rights are enforced separately from rank.

Human-in-the-loop approvals — six guarantees

1. Two people, always	Proposers and approvers are different roles; no one decides their own proposal.
2. Approve exactly what was proposed	The proposal is SHA-256 pinned; if it changes, the approval is refused.
3. Nothing is silent	A written justification to propose; a written reason to approve, reject or withdraw.
4. Compliance first	Vigil screens every proposal; prohibited demographic targeting is blocked outright.
5. Decisions expire	Stale proposals move to Expired and cannot be approved.
6. Everything on the record	Each state change writes a hash-chained audit entry.

Audit, privacy and observability

- **Tamper-evident audit ledger:** every answer and decision is hashed and chained to the previous entry; the chain survives restarts and can be verified end to end.
- **Privacy:** victim identities in sexual offences and cases involving children are redacted on every answer (BNS §72, POCSO §23); personal data is redacted before any text reaches the cloud model (DPDP Act 2023 principles); no individual-level predictions.
- **Explainability:** citations to FIRs, evidence class per claim, a reasoning trail per answer, and a chain-of-custody strip.
- **Observability:** per-answer traces and spans, a trust dashboard, and an evaluation set; the test suite (903 tests) includes 51 contract tests that fail the build if Antar breaks its own rules.
- **Verify it live:** audit chain integrity at /api/audit/integrity (Admin) · trust dashboard at /trust · traces at /api/obs/dashboard · how any figure was derived at /api/derivation/{id}.
- **Standards alignment:** ICD 203 analytic standards, BLUF, NIST AI RMF and EU AI Act principles for high-risk law-enforcement AI (human oversight, logging, accuracy, transparency).

12 Requirements coverage and measured impact

Against the Datathon solution framework. Status is stated honestly.

Framework requirement	Status	How Antar meets it
1 Conversational interface (EN + KN, follow-ups, voice, PDF export)	Delivered	Chat in both languages, session context, voice in and out, printable session report
2 Network & relationship analysis	Delivered	Network agent with centrality and communities; shared identifiers across districts (T19–T21)
3 Pattern & trend analytics	Delivered	Trends, rankings, beat hotspots, crime-cycle seasonality, Gi* clusters
4 Sociological insights	Partial	Age / gender demographics (T7); socio-economic indicators are not in the FIR data
5 Criminological profiling & risk	Partial	Repeat offenders, accused history, profiles; modus-operandi text is sparse in the data
6 Investigator decision support	Delivered	Case lists, K/D/I/U dossiers, similar-case linkage, devil's-advocate leads
7 Financial crime & money trails	Partial	UPI ID / IP / vehicle links across districts; transaction data is limited
8 Forecasting & early warning	In progress	Anomaly scan, Gi*, conformal ranges live; beat-level forecast with measured accuracy being finalised
9 Explainable & transparent AI	Delivered	Citations, evidence classes, reasoning trail, custody strip, model and agent traces
10 Role-based access & governance	Delivered	Six roles, HITL, hash-chained audit, privacy redaction, observability

What changed for the officer

Before	With Antar
A records request and a wait	An answer in under a second, with its FIRs
A confident number that may answer a different question	A refusal with the nearest valid options
English dashboards	Kannada text and voice, same logic as English
Charts with no provenance	Every claim labelled Known, Derived, Inferred or Unknown
Actions taken without a record	Two-person approval on a tamper-evident ledger

13 Final-day kit

Run of show (15–20 min)

Min	Beat	Min	Beat
0–2	Problem and who it's for	9–11	Brain: 10 lakh FIRs, crime-cycle seasonality
2–4	Kidnapping ranking — bottom line, badge, followed-by	11–13	Network: shared identifiers across districts
4–6	Kannada: ನಿನ್ನೆ ಕಳ್ಳತನ ಎಷ್ಟು + Listen	13–15	Map / Predict: hotspots with reasons
6–8	Refusals: 'burgalry'; absconding in Indiranagar	15–18	Propose patrol → Approver approves → audit chain
8–9	Expert console: evidence by claim, trace	18–20	Close: four laws, 903 tests, scale path

Before going on stage

- Open the deployed link 15 minutes early (warms the model and caches); sign in with both demo accounts in separate windows (Operations, Approver).
- Ask each demo question once in advance — answers and spoken audio are then cached and instant.
- For arrest / absconding questions, use Karnataka districts (Bengaluru City, Mysuru City, Belagavi Dist).

Likely judge questions

How do you stop hallucination?	The model never writes SQL or picks numbers; 19 reviewed templates, resolve-or-refuse, evidence classes — enforced by tests.
What if the cloud model is down?	The deterministic writer answers every template in both languages; the LLM only improves wording.
Can it scale to crores of FIRs?	Covering indexes, cached aggregates and cubes; ZCQL rendering for Catalyst Data Store; local GPU inference.
Is the forecast accurate?	Ranges come from split-conformal intervals; beat-level accuracy is being measured with the Predictive Accuracy Index.
Privacy?	Victim-identity redaction (BNS §72, POCSO §23), PII stripped before the cloud, no individual predictions, two-person approval.

Honest notes for Q&A

- The real KSP extract ends **10 March 2024**; later rows and some sub-zones (e.g. Indiranagar, Rajajinagar) are demonstration data without arrest counts — Antar says so when asked.
- Modus-operandi text, relationship and financial links are thin in the source extract; the pipelines are built for a full CCTNS feed.
- The Predict screen's model card is still a fixed description and will be regenerated from the forecast back-test — avoid opening it on stage until then.
- Roadmap: beat-level forecast with measured accuracy · CCTNS integration · local GPU model · full-text search · Catalyst deployment hardening.